

Energy-Energy Correlator from the AdS Virasoro-Shapiro Amplitude

Lecheng Ren

Queen Mary University of London

Prague Spring Amplitudes, May 13, 2026

Based on arXiv:2601.05312 with Bo Wang and Congkao Wen
WIP with Max Jackson, Bo Wang, and Congkao Wen

Why energy correlators

- measured in experiments (e^+e^- Colliders, ep Colliders, pp Colliders, ...)
- Infrared safe; bridge formal theory and experiments; ...

[Moult, Zhu 2025 review]

Why $\mathcal{N} = 4$ super Yang-Mills (SYM)

- AdS/CFT, allow for analytic computation at strong coupling, exact results

- At weak coupling, there is fruitful knowledge at high multiplicity and high loop order.

[Dixon, Luo, Shtabovenko, Yang, Zhu; Henn, Sokatchev, Yan, Zhiboedov; He, Jiang, Yang, Zhang; Volovich, Wu, Yan;...]

- Strong coupling: the 4-point AdS amplitude is known up to second curvature correction.

[Alday, Hansen, Silva; Aprile, Drummond, Paul, Santagata; Abl, Heslop, Lipstein; Rastelli, Zhou]

- In this talk, I describe a direct algorithm for obtaining the 2-point energy correlator in $\mathcal{N} = 4$ SYM from AdS amplitudes
- Reformulate in worldsheet representation

- Correlation functions of detector operators:

$$\text{EEC}(\xi) = \langle \Psi | \mathcal{E}(\mathbf{n}_1) \mathcal{E}(\mathbf{n}_2) | \Psi \rangle .$$

$$\mathcal{E}(\mathbf{n}) = \frac{1}{4} \int_{-\infty}^{\infty} du \lim_{r \rightarrow \infty} r^2 T_{\mu\nu}(u+r, r\mathbf{n}) \bar{n}^\mu \bar{n}^\nu ,$$

- Energy + momentum conservation: The angular dependence is naturally expanded in Legendre multipoles (use $\xi = \sin^2 \frac{\theta}{2}$):

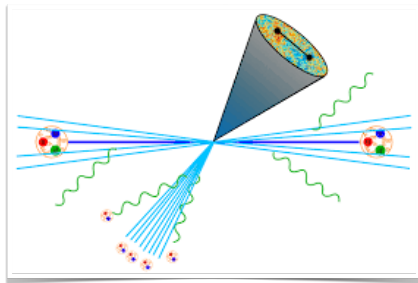
$$\int_0^1 d\xi \text{EEC}(\xi) = 1, \quad \int_0^1 d\xi (1 - 2\xi) \text{EEC}(\xi) = 0.$$

$$\Rightarrow \boxed{\text{EEC}(\xi) = \sum_{s=0}^{\infty} c_s P_s(1 - 2\xi)}, \quad c_0 = 1, \quad c_1 = 0.$$

- Energy flux operator

[Sternman, 1975]

$$\mathcal{E}(\hat{n})|X\rangle = \sum_{k \in X} k^0 \delta^2(\Omega_n - \Omega_k) |X\rangle.$$



Energy correlators provide a link between measurements of energy flux at macroscopic scales and local correlators underlying microscopic dynamics

- The averaged null energy condition (ANEC)

$$\langle X|\mathcal{E}|X\rangle \geq 0$$

Related to three-point functions of local operators (with one stress tensor): conformal collider bounds in the case of CFTs

[Hofman, Maldacena]

We'll be interested in two-point functions: energy-energy correlators

Examples of EEC

- Weak coupling:

$$\text{EEC}(\xi) = \frac{1}{2}(\delta(\xi) + \delta(1 - \xi)) + O(\lambda).$$

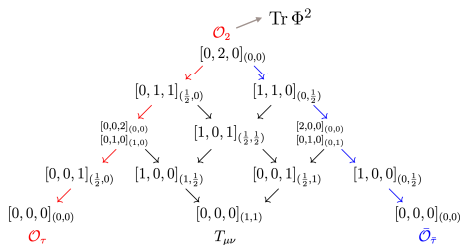
- Strong coupling:

$$\text{EEC}(\xi) = 1 + O(1/\lambda).$$

Analytic calculation in the strong coupling regime made possible due to AdS/CFT correspondence

[Hofman, Maldacena]

$\mathcal{N} = 4$ stress tensor multiplet



EECs in $\mathcal{N} = 4$ SYM are related to

$$\langle \mathcal{O}_p \mathcal{O}_2 \mathcal{O}_2 \mathcal{O}_p \rangle$$

Superconformal primary operators

$$\mathcal{O}_p(x, Y) = \text{Tr}(\Phi^{I_1}(x) \cdots \Phi^{I_p}(x)) Y_{I_1} \cdots Y_{I_p}$$

- $\Phi^I(x)$ are scalars, and Y_I are $\text{SO}(6)$ R-symmetry null vectors.
- They are 1/2 BPS operators.

- We will be interested in correlators in planar $\mathcal{N} = 4$ SYM of the form

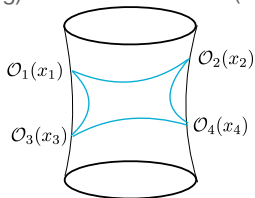
$$\langle \mathcal{O}_p(x_1, Y_1) \mathcal{O}_2(x_2, Y_2) \mathcal{O}_2(x_3, Y_3) \mathcal{O}_p(x_4, Y_4) \rangle$$

- Superconformal Ward identity

[Eden, Petkou, Schubert, Sokatchev; Nirschl, Osborn]

$$\begin{aligned} & \langle \mathcal{O}_p(x_1, Y_1) \mathcal{O}_2(x_2, Y_2) \mathcal{O}_2(x_3, Y_3) \mathcal{O}_p(x_4, Y_4) \rangle \\ &= \mathcal{G}_{\text{free}}(x_i, Y_i) + \underbrace{\mathcal{I}_4(x_i, Y_i)(d_{14})^{p-2}}_{\text{Fixed by symmetry}} \underbrace{\mathcal{T}_p(u, v)}_{\text{Our focus}} \end{aligned}$$

String amplitudes in AdS $\longleftrightarrow$ Correlators in the dual CFT
 (Type IIB superstring) ($\mathcal{N}=4$ super Yang-Mills)



$$A(s, t; \alpha'/R^2) \leftrightarrow \langle \mathcal{O}_1(x_1) \mathcal{O}_2(x_2) \mathcal{O}_3(x_3) \mathcal{O}_4(x_4) \rangle \quad 1/\sqrt{\lambda} = \alpha'/R^2$$

AdS radius



Correlators in Mellin space

[Mack; Penedones]

$$\begin{aligned}
\mathcal{T}_p(u, v) = & \frac{1}{4p} \int_{-i\infty}^{i\infty} \frac{ds dt}{(2\pi i)^2} u^{\frac{1}{2}(\tilde{u}-p)} v^{\frac{1}{2}(s-p)} \\
& \times \Gamma^2\left(\frac{p+2-s}{2}\right) \Gamma\left(2-\frac{t}{2}\right) \Gamma\left(p-\frac{t}{2}\right) \\
& \times \Gamma^2\left(\frac{p+2-\tilde{u}}{2}\right) \mathcal{M}_p(s, t).
\end{aligned}$$

with Mellin variables

$$s + t + \tilde{u} = 2p.$$

for the simplest case $p = 2$,

$$s + t + \tilde{u} = 4.$$

At strong coupling (or low-energy) expansion

$$\mathcal{M}_p(s, t) = \underbrace{\frac{4p}{\Gamma(p-1)} \frac{1}{(s-p)(t-2)(p-s-t)}}_{\text{supergravity}} + \underbrace{\frac{\Gamma(4+p)}{\Gamma(p)\Gamma(p-1)} \frac{\zeta(3)}{\lambda^{3/2}}}_{\substack{\text{stringy corrections} \\ \text{polynomials}}} + \dots$$

- The EECs are Wightman functions: Euclidean to Wightman

[Mack, 2009]

$$x_{ij}^2 \longrightarrow -x_{ij}^2 + i\epsilon x_{ij}^0 \theta(j-i)$$

Mellin amplitudes make analytic continuation relatively easier

- After the Wick rotation and Fourier transform, the EECs can be expressed as

[Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov]

$$\text{EEC}_p(\xi) = \int_{-i\infty}^{i\infty} \frac{ds dt}{(2\pi i)^2} \mathcal{M}_p(s, t) K_p(t, \xi).$$

- We start from the four-point correlator of the half-BPS operator

$$\mathcal{O}_{20'}(x, y) = y^I y^J \text{Tr} \left[\Phi^I \Phi^J \right] (x), \quad y \cdot y = 0.$$

- After the light-ray transform, this gives the following integral formula:

[Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov]

$$\text{EEC}(\xi) = \int \frac{ds dt}{(2\pi i)^2} M(s, t) K(t, \xi),$$

$$K(t, \xi) = \frac{-\pi t^2 (t-2)^2}{128 \xi^3 \sin(\pi t/2)} \left(\frac{\xi}{1-\xi} \right)^{1+t/2}.$$

- $M(s, t)$: Mellin amplitude of the four-point correlation function.

$$M(s, t) = \underbrace{\frac{8}{(s-2)(t-2)(\tilde{u}-2)}}_{\text{SUGRA}} + \underbrace{\frac{120\zeta_3}{\lambda^{3/2}}}_{\text{first stringy correction}} + (\text{polynomials in } s, t)$$

$$\text{EEC}(\xi) = \int \frac{ds dt}{(2\pi i)^2} M(s, t) K(t, \xi).$$

- SUGRA:

$$\text{EEC}(\xi) = 1.$$

- Stringy corrections: expand the Mellin amplitude in powers of t . Since

$$Q_n(\xi) = \int \frac{dt}{2\pi i} t^n K(t, \xi).$$

We have

$$\text{EEC}(\xi) = 1 + \sum_{k \geq 0} \lambda^{-k/2-1} \sum_{n \geq 0} c_{k,n} Q_n(\xi),$$

and Q are reorganization of Legendre polynomials

$$Q_0(\xi) = P_2(1 - 2\xi), \quad Q_1(\xi) = P_2(1 - 2\xi) + 3P_3(1 - 2\xi).$$

[Belitsky, Hohenegger, Korchemsky, Sokatchev, Zhiboedov;
Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; ...]

- The s -integral, which determines coefficients $c_{k,n}(p)$, however is not well-defined for low-energy expansion of Mellin amplitudes
- The light transform and low-energy (strong coupling) expansion do not commute
- It requires an alternative organising principle: AdS Virasoro-Shapiro amplitudes that resum low-energy expansions

e.g. flat space

$$A^{(0)}(S, T) = -\frac{\Gamma(-S)\Gamma(-T)\Gamma(S+T)}{\Gamma(S+1)\Gamma(T+1)\Gamma(1-S-T)}.$$

[Virasoro; Shapiro]

- AdS curvature expansion

[Alday, Hansen]

$$A(S, T) = \sum_{k=0}^{\infty} \lambda^{-k/2} A^{(k)}(S, T), \quad \frac{1}{\sqrt{\lambda}} = \frac{\alpha'}{R^2}.$$

- Relate to Mellin amplitudes via a Borel transform

[Penedones]

$$\mathcal{M}_p(\hat{s}, \hat{t}) = \frac{1}{2\lambda^{3/2}\Gamma(p-1)\Gamma(p)} \int_0^\infty d\beta \beta^{p+3} e^{-\beta} A(\beta\hat{s}, \beta\hat{t}).$$

with

$$\hat{s} = \frac{s_1}{2\sqrt{\lambda}} = \frac{s - 2p/3}{2\sqrt{\lambda}},$$

$$\hat{t} = \frac{s_2}{2\sqrt{\lambda}} = \frac{t - 2p/3}{2\sqrt{\lambda}}.$$

World-sheet formulation for AdS VS amplitudes

[Alday, Hansen, Silva]

- The flat space amplitude

$$A^{(0)}(S, T) = \int d^2z |z|^{-2S-2} |1-z|^{-2T-2}$$

- AdS curvature corrections

$$A^{(k)}(S, T) = \int d^2z |z|^{-2S-2} |1-z|^{-2T-2} \mathcal{G}_{\text{tot}}^{(k)}(z, \bar{z}; S, T)$$

$\mathcal{G}_{\text{tot}}^{(k)}(z, \bar{z}; S, T)$:

single-valued multiple polylogarithms (SVMP) with weight $3k$.

Use the Borel representation inside the EEC integral

[Penedones]

$$\text{EEC}(\xi) = \frac{1}{\lambda} \int \frac{dt}{2\pi i} \int_0^\infty d\beta e^{-\beta} \beta^4 I(\beta \hat{t}) K(t, \xi) \quad \hat{s} = \frac{s - \frac{4}{3}}{2\sqrt{\lambda}}, \quad \hat{t} = \frac{t - \frac{4}{3}}{2\sqrt{\lambda}}.$$

$$I(T) = \int \frac{dS}{2\pi i} A(S, T). \quad S = \beta \hat{s}, \quad T = \beta \hat{t}.$$

$A(T)$ has a worldsheet formula

$$A(S, T) = \int_c d^2 z |z|^{-2S-2} |1-z|^{-2T-2} G_{\text{tot}}(z, \bar{z}; S, T).$$

[Alday, Hansen, Silva; see also Schlotterer's talk]

Use the Borel representation inside the EEC integral

[Penedones]

$$\text{EEC}(\xi) = \frac{1}{\lambda} \int \frac{dt}{2\pi i} \int_0^\infty d\beta e^{-\beta} \beta^4 I(\beta \hat{t}) K(t, \xi) \quad \hat{s} = \frac{s - \frac{4}{3}}{2\sqrt{\lambda}}, \quad \hat{t} = \frac{t - \frac{4}{3}}{2\sqrt{\lambda}}.$$

$$I(T) = \int \frac{dS}{2\pi i} A(S, T). \quad S = \beta \hat{s}, \quad T = \beta \hat{t}.$$

$A(T)$ has a worldsheet formula

$$A(S, T) = \int_c d^2 z |z|^{-2S-2} |1-z|^{-2T-2} G_{\text{tot}}(z, \bar{z}; S, T).$$

[Alday, Hansen, Silva; see also Schlotterer's talk]

- G_{tot} : svMPL of weight $3k$; singularity at most $1/S^2$, $1/T^2$, $1/U^2$.
- T and β integral: straightforward after Taylor expansion over T in the integrand.

Use the Borel representation inside the EEC integral

[Penedones]

$$\text{EEC}(\xi) = \frac{1}{\lambda} \int \frac{dt}{2\pi i} \int_0^\infty d\beta e^{-\beta} \beta^4 I(\beta \hat{t}) K(t, \xi) \quad \hat{s} = \frac{s - \frac{4}{3}}{2\sqrt{\lambda}}, \quad \hat{t} = \frac{t - \frac{4}{3}}{2\sqrt{\lambda}}.$$

$$\boxed{I(T) = \int \frac{dS}{2\pi i} A(S, T).} \quad S = \beta \hat{s}, \quad T = \beta \hat{t}.$$

$A(T)$ has a worldsheet formula

$$A(S, T) = \int_C d^2 z |z|^{-2S-2} |1-z|^{-2T-2} G_{\text{tot}}(z, \bar{z}; S, T).$$

[Alday, Hansen, Silva; see also Schlotterer's talk]

- G_{tot} : svMPL of weight $3k$; singularity at most $1/S^2$, $1/T^2$, $1/U^2$.
- T and β integral: straightforward after Taylor expansion over T in the integrand.
- Essential object to compute is $I(T)$.

Use the Borel representation inside the EEC integral

[Penedones]

$$\text{EEC}(\xi) = \frac{1}{\lambda} \int \frac{dt}{2\pi i} \int_0^\infty d\beta e^{-\beta} \beta^4 I(\beta \hat{t}) K(t, \xi) \quad \hat{s} = \frac{s - \frac{4}{3}}{2\sqrt{\lambda}}, \quad \hat{t} = \frac{t - \frac{4}{3}}{2\sqrt{\lambda}}.$$

$$\boxed{I(T) = \int \frac{dS}{2\pi i} A(S, T).} \quad S = \beta \hat{s}, \quad T = \beta \hat{t}.$$

$A(T)$ has a worldsheet formula

$$A(S, T) = \int_c d^2 z |z|^{-2S-2} |1-z|^{-2T-2} G_{\text{tot}}(z, \bar{z}; S, T).$$

[Alday, Hansen, Silva; see also Schlotterer's talk]

- G_{tot} : svMPL of weight $3k$; singularity at most $1/S^2$, $1/T^2$, $1/U^2$.
- T and β integral: straightforward after Taylor expansion over T in the integrand.
- Essential object to compute is $I(T)$.

trick: change the order of string moduli and S integral

For terms of the form

$$J_j(T) = \int_{\varepsilon^+ - i\infty}^{\varepsilon^+ + i\infty} \frac{dS}{2\pi i} \int d^2z S^j |z|^{-2S-2} |1-z|^{-2T-2} J(z, \bar{z}; T),$$

the S integral localizes the world-sheet integral:

$$j < 0 : \quad \text{contour deformation} \quad \Longrightarrow \quad \boxed{|z| < 1},$$

$$j \geq 0 : \quad \delta^{(j)}(\log |z|^2) \quad \Longrightarrow \quad \boxed{|z| = 1}.$$

reduce EEC to simple boundary/disk integral over moduli space!

- ① Expand the AdS moduli space integrand in powers of $\lambda^{-1/2}$.
- ② Do the S contour integral first.
- ③ Reduce to disk and boundary-circle integrals in z .
- ④ The β and T integrals are straightforward.

The leading term (flat space contribution): only the disk part

$$\begin{aligned} I^{(0)}(T) &= \int_{|z|<1} d^2z |1-z|^{-2T-2} (-2 \ln |z|) \\ &= \frac{1}{T^2} \left[\frac{\Gamma(1-2T)}{\Gamma(1-T)^2} - 1 \right] \end{aligned}$$

Small T -expansion:

$$I^{(0)}(T) = \zeta_2 + 2\zeta_3 T + \frac{19}{4}\zeta_4 T^2 + \dots$$

This gives

$$c_{0,0} = 24\zeta_2 = 4\pi^2, \quad c_{1,0} = 120\zeta_3, \quad c_{2,0} = 855\zeta_4.$$

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Wang, Wen, LR]

First curvature correction:

$$\begin{aligned} I^{(1)}(T) &= \int_{|z|<1} d^2z |z|^{2T-2} |1-z|^{-2T-2} \log^3 \frac{|z|^2}{|1-z|^2} \\ &\quad + \oint_{|z|=1} d\varphi |1-z|^{-2T-2} (\text{Li}_3(z) \oplus \log^3(1-z) \oplus \zeta_3) \\ &= \frac{1}{3T^4} \frac{d}{dT} \left[T^5 \frac{d^2}{dT^2} \frac{\Gamma(-2T)}{\Gamma(1-T)^2} \right] + 2\zeta_3 \frac{\Gamma(-2T-1)}{\Gamma(-T)^2} F(T) - F(T-1) \end{aligned}$$

Small- T expansion

$$I^{(1)}(T) = \zeta_2 - \frac{10}{3}\zeta_3 + \left(-2\zeta_2 + 4\zeta_3 - \frac{53}{2}\zeta_4 \right) T + \dots$$

Therefore

$$c_{1,1} = 24\zeta_2 - 240\zeta_3, \quad c_{2,1} = -120\zeta_2 + 240\zeta_3 - 3870\zeta_4.$$

[Alday, Hansen, Silva; Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Wang, Wen, LR]

Examples from 1st curvature correction

[Alday, Hansen; Fardelli, Hansen, Silva]

• disc integral

$$\begin{aligned}
 I_1^{(1)}(T) &= \frac{p^2 T}{36} \int_{|z| < 1} d^2 z |z|^{2T-2} |1 - z|^{-2T-2} \log^3 \frac{|z|^2}{|1 - z|^2} \\
 &= -\frac{p^2 T}{36} \frac{\partial^3}{\partial T^3} \frac{\Gamma(-2T)}{\Gamma(1 - T)^2}.
 \end{aligned}$$

• circle integral

$$\begin{aligned}
 I_3^{(1)}(T) &= \oint_{|z|=1} \frac{d\theta}{2\pi} |1 - z|^{-2T-2} (2\zeta(3) - \text{Li}_3(z) - \text{Li}_3(\bar{z})) \\
 &= \frac{2\zeta(3)\Gamma(-2T - 1)}{\Gamma(-T)^2} + F(T).
 \end{aligned}$$

2nd curvature involves weight-6 SVMP and is only known for $p = 2$

[Alday, Hansen]

To determine the EEC to next order, we actually only need $I^{(2)}(T)|_{T=0}$

- This term receives contribution only from the unit circle integral, which greatly simplifies the calculation (even though still not easy)

$$I^{(2)}(T) = 6\zeta(2) - \frac{13}{2}\zeta(3) + \frac{263}{6}\zeta(4) + \zeta(2)\zeta(3) - \frac{5}{3}\zeta(5) + O(T).$$

- Putting together flat-space, 1st and 2nd curvatures, we determine the EEC up to λ^{-2}

$$\text{EEC}(\xi) = 1 + \frac{24\zeta_2}{\lambda} Q_0(\xi) + \frac{1}{\lambda^{3/2}} [(24\zeta_2 - 240\zeta_3)Q_0(\xi) + 120\zeta_3 Q_1(\xi)] + \cdots$$

Using

$$Q_0 = P_2(1 - 2\xi), \quad Q_1 = P_2(1 - 2\xi) + 3P_3(1 - 2\xi),$$

we get the Legendre moments

$$c_2 = \frac{4\pi^2}{\lambda} + \frac{4\pi^2 - 120\zeta_3}{\lambda^{3/2}} + \cdots, \quad c_3 = \frac{360\zeta_3}{\lambda^{3/2}} + \cdots.$$

[Wang, Wen, LR]

The EEC to the highest order we have so far

$$\begin{aligned} \text{EEC}_2(\xi) = 1 + \frac{c_{0,0}}{\lambda} Q_{0,2}(\xi) + \frac{c_{1,1} Q_{0,2}(\xi) + c_{1,0} Q_{1,2}(\xi)}{\lambda^{3/2}} \\ + \frac{c_{2,2} Q_{0,2}(\xi) + c_{2,1} Q_{1,2}(\xi) + c_{2,0} Q_{2,2}(\xi)}{\lambda^2} + O(\lambda^{-5/2}). \end{aligned}$$

$$c_{0,0} = 24\zeta(2), \quad c_{1,0} = 120\zeta(3), \quad c_{2,0} = 855\zeta(4),$$

$$c_{1,1} = 24\zeta(2) - 240\zeta(3),$$

$$c_{2,1} = -120\zeta(2) + 240\zeta(3) - 3870\zeta(4),$$

$$c_{2,2} = 304\zeta(2) - 476\zeta(3) + 24\zeta(2)\zeta(3) + 4692\zeta(4) - 40\zeta(5).$$

[Hofman, Maldacena; Jackson, Wang, Wen, LR]

It's instructive to reproduce the results from a “traditional” method:
low-energy expansion

The low-energy expansion gives

$$M(\hat{s}, \hat{t}) = M^{\text{SUGRA}} \frac{1}{\lambda^{3/2}} \sum_{\ell \geq 0} \lambda^{-\ell/2} M^{(\ell)}(\hat{s}, \hat{t}).$$

[Alday, Hansen, Silva]

If

$$M^{(\ell)}(\hat{s}, \hat{t}) = \sum_j M_j^{(\ell)}(\hat{s}) \hat{t}^j, \quad \hat{c}_{j,\ell} = \int \frac{d\hat{s}}{2\pi i} M_j^{(\ell)}(\hat{s}),$$

then

$$c_{k,0} = 2^{-k} \hat{c}_{k,0}, \quad c_{k+1,1} = 2^{-k} \left(\hat{c}_{k,1} - \frac{2(k+1)}{3} \hat{c}_{k+1,0} \right).$$

For $k = 0$:

$$c_{1,1} = \hat{c}_{0,1} - \frac{2}{3} \hat{c}_{1,0} = 24\zeta_2 - 240\zeta_3.$$

[Wang, Wen, LR]

- The first curvature contribution

[Alday, Hansen, Silva]

$$\begin{aligned}\alpha_{a,0}^{(1)} &= 2\zeta(2a+1)\zeta(3) - 2\zeta(2a+1,3) - (2a+1)\zeta(2a+2,2) \\ &\quad + \frac{1}{3}(6a^2 + 23a + 3)\zeta(2a+3,1) \\ &\quad - \frac{1}{3}(4a^3 + 16a^2 + 13a + 6)\zeta(2a+4).\end{aligned}$$

- Setting $a = -1/2$ is straightforward, except one term

$$\lim_{\varepsilon \rightarrow 0} -(2a+1)\zeta(2a+2,2) \Big|_{a \rightarrow -\frac{1}{2} + \frac{\varepsilon}{2}} = \lim_{\varepsilon \rightarrow 0} -\varepsilon \zeta(1+\varepsilon, 2) = -\zeta(2).$$

we included this term, but not by

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov]

Both calculations compared below are already performed at first curvature order.

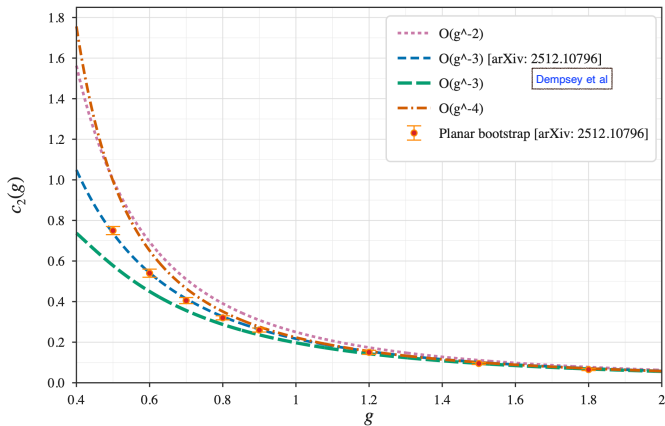
- The noncommuting operations are the infinite mass-level sum and the EEC S -integral:

$$\int_{-i\infty}^{+i\infty} \frac{dS}{2\pi i} \sum_{\delta=1}^{\infty} A_{\delta}^{(1)}(S, 0) \neq \sum_{\delta=1}^{\infty} \int_{-i\infty}^{+i\infty} \frac{dS}{2\pi i} A_{\delta}^{(1)}(S, 0).$$

$$\Delta I^{(1)}(0) = -\zeta(2), \quad \Delta c_2|_{\lambda^{-3/2}} = -24\zeta(2) = -4\pi^2.$$

It does not by itself prove that the latter equals the coefficient obtained by first performing the EEC transform at finite λ and only then expanding at strong coupling.

- Wilson coefficient for 2nd curvature is much more involved (not known in the literature)
- Setting $a = -1/2$ is divergent term by term, so we have to introduce a regulator
- All the $1/\varepsilon$ divergences cancel out, and the finite part agrees with world-sheet calculation



$$g^2 = \lambda/(16\pi^2)$$

[Zhiboedov et al.; Jackson, Wang, Wen, LR]

- Can we compute EECs at strong coupling directly without using AdS VS amplitudes? this may also help to resolve the discrepancy
- EECs with heavy states, exploiting results from heavy-state (or large charge) expansion.
- Modular completion due to S-duality: one possible starting point is the leading stringy correction
- Less supersymmetric theories

- We computed the EEC of $\mathcal{N} = 4$ SYM from AdS worldsheet integral.
- Represent EEC into integral over the moduli space, the computation simplifies.
- For sanity check, we compare with the result from low energy expansion of Mellin amplitudes.
- bootstrapp moduli space integrand for EEC?
- higher curvature expansion? higher genus?

Thank you!

Complementary pages

- The Wang, Wen, LR Appendix C check differs from the corresponding calculation quoted in Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov
- The difference comes from a term in Alday's low-energy expansion,

$$(2a + 1)\zeta(2a + 2),$$

which is lost when we do summation over massive pole in OPE in the final step.

$$[\text{Wang, Wen, LR}]: \quad c_2|_{\lambda^{-3/2}} = 4\pi^2 - 120\zeta_3,$$

$$[\text{DKPZZ}]: \quad c_2|_{\lambda^{-3/2}} = 8\pi^2 - 120\zeta_3.$$

[Alday, Hansen, Silva; Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Wang, Wen, LR]

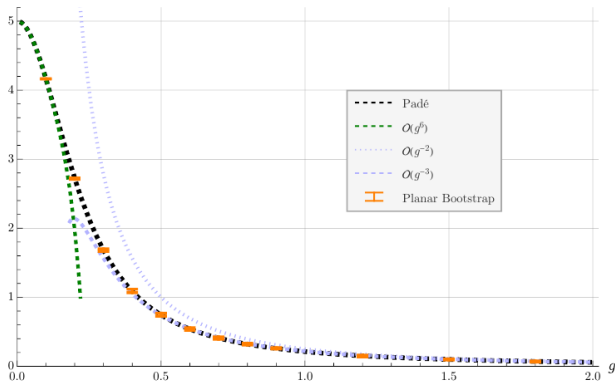
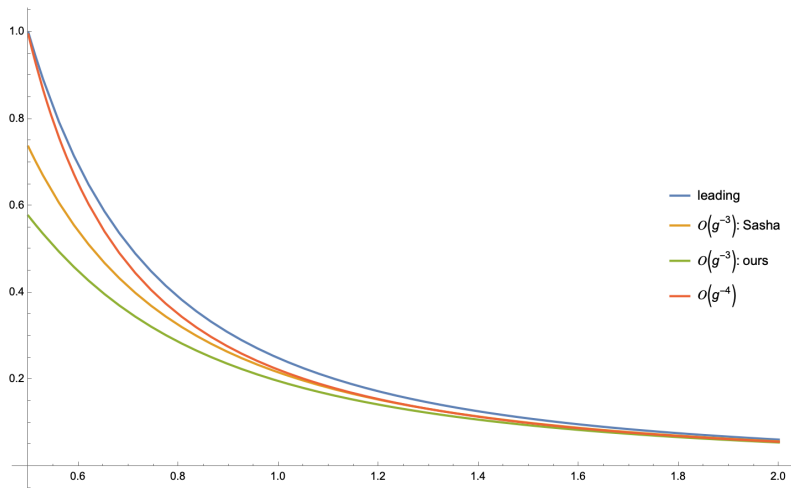


Figure 2. In orange we plot the bootstrap bounds on $c_2(g)$ in the planar limit, as obtained in [Section 6.3](#). We also plot the perturbative 3-loop result (dashed green), the leading strong coupling result (dotted blue), as well as the strong coupling result (4.6) up to subleading order in the strong coupling expansion (dashed blue). In black we plot a Padé approximation based on the 3-loop result at weak coupling and the leading strong coupling result. In particular, the subleading strong prediction (4.6) (in dashed blue) improves the agreement with the bootstrap bounds significantly relative to the leading result (in dotted blue).

[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov]



[Dempsey, Karlsson, Pufu, Zahraee, Zhiboedov; Jackson, Wang, Wen, LR]